



COLLEGE
OF
MEDICINE

Urinary Incontinence

Dr Dalya Dler Nuri
MBChB, DGO, GBCOG



COLLEGE
OF
MEDICINE

objectives

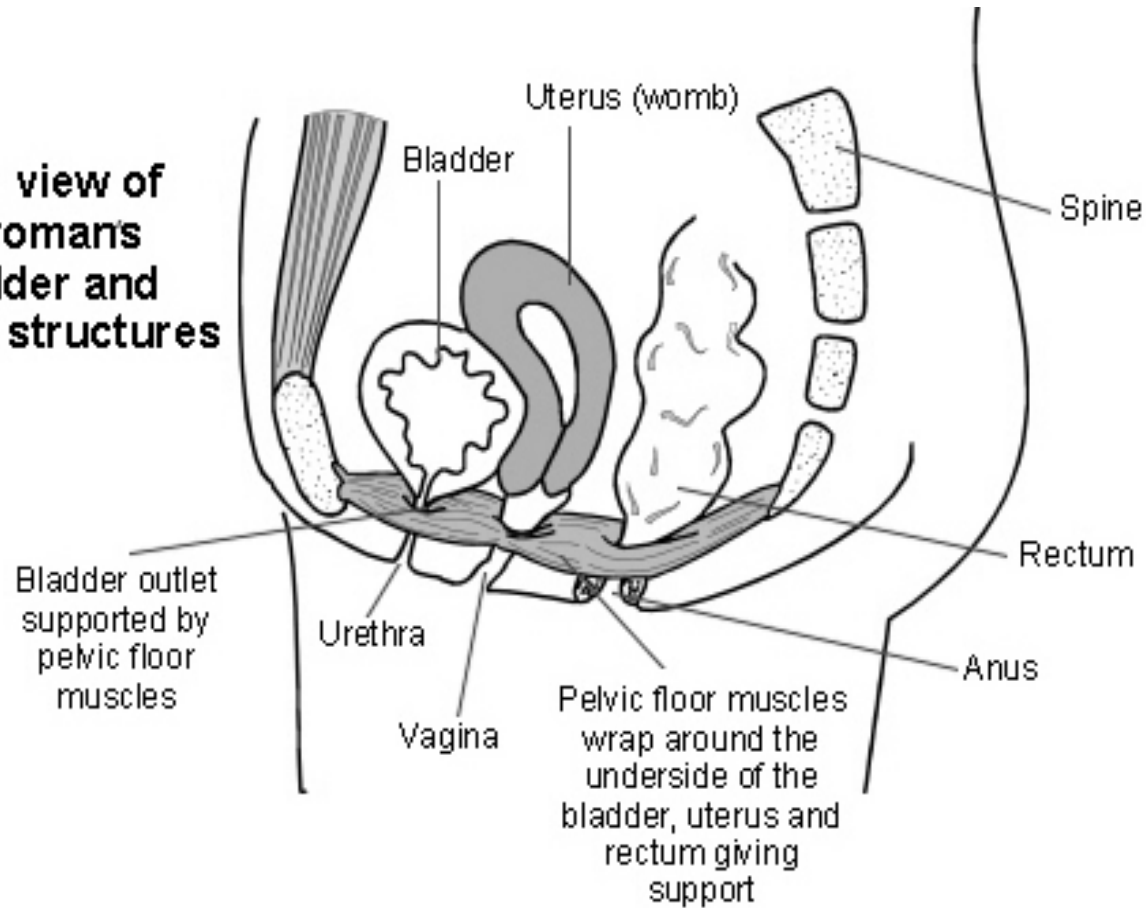
- Know the definition and types of urinary incontinence .
- Learn how to assess the patient with incontinence by means of history, examination and relevant investigations.
- Understand the principles of urodynamic testing
 - Understand the principles of treatment of incontinence and be able to describe the effectiveness of each treatment, together with an understanding of potential side-effects and complications
-



Introduction

- Urinary incontinence (UI) is a common symptom that can affect women of all ages, with a wide range of severity and nature.
- UI is defined by the International Continence Society as 'the complaint of any involuntary leakage of urine .
- Stress UI is involuntary urine leakage on effort or exertion or on sneezing or coughing.
- Urgency UI is involuntary urine leakage accompanied by urgency (sudden compelling desire to urinate that is difficult to delay).
- Mixed UI is involuntary urine leakage associated with both urgency and exertion, effort, sneezing or coughing.
- Overactive bladder (OAB) is defined as urgency that occurs with or without urgency UI and usually with frequency and nocturia. OAB that occurs with incontinence is known as 'OAB wet'. OAB that occurs without incontinence is known as 'OAB dry'.

**Side view of
a woman's
bladder and
related structures**





- History-taking and physical examination
At the initial clinical assessment, categorise the woman's urinary incontinence (UI) as stress UI (SUI), mixed UI, or urgency UI/overactive bladder (OAB). Start initial treatment on this basis.
- In mixed UI, direct treatment towards the predominant symptom.
- During the clinical assessment seek to identify relevant predisposing and precipitating factors for additional investigation and treatment.

Assessment of pelvic floor muscles

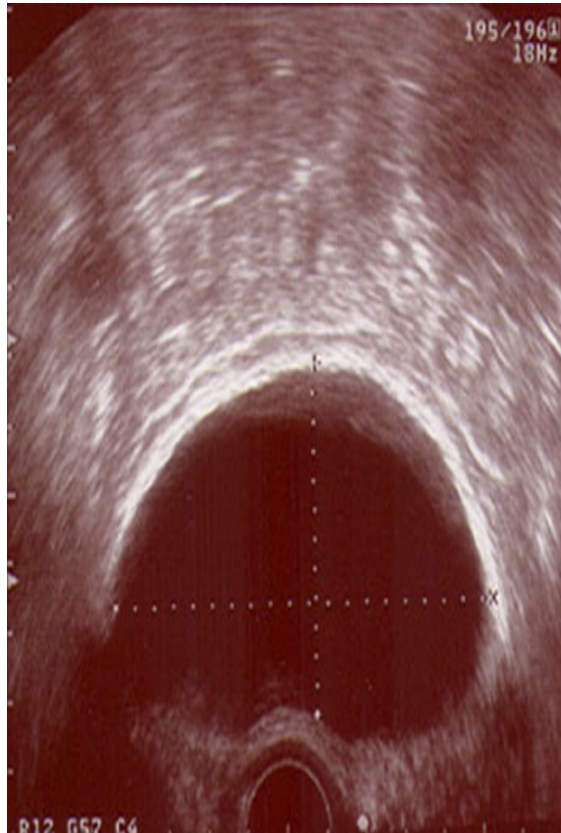
- Undertake routine digital assessment to confirm pelvic floor muscle contraction .
- Assessment of prolapse
- Urine testing
Undertake a urine dipstick test in all women presenting with UI to detect the presence of blood, glucose, protein, leucocytes and nitrites in the urine.
- If women have symptoms of urinary tract infection (UTI) and their urine tests positive for both leucocytes and nitrites send a midstream urine specimen for culture and analysis of antibiotic sensitivities. If women have symptoms of UTI and their urine tests negative for either leucocytes or nitrites send a midstream urine specimen for culture and analysis of antibiotic sensitivities.



- Assessment of residual urine
Measure post-void residual volume by bladder scan or catheterisation in women with symptoms suggestive of voiding dysfunction or recurrent UTI. .
- Bladder diaries
Use bladder diaries in the initial assessment of women with UI or OAB. Encourage women to complete a minimum of 3 days of the diary covering variations in their usual activities, such as both working and leisure days.
- Pad testing
Do not use pad tests in the routine assessment of women with UI.



COLLEGE
OF
MEDICINE





COLLEGE OF MEDICINE

Time	Type of drink	Amount drunk (ml)	Urine passed (ml)	had a leak Did you experience urgency? Were you coughing, sneezing, exercising or laughing?	
					Cross if you changed your pad
7-8 am	Coffee	100	200	X urgency	X
8-9 am			30		

Urodynamic testing

- The purpose of urodynamic testing is to reproduce a micturition cycle (bladder filling and voiding) while recording abdominal and bladder pressure and attempting to reproduce the patient's symptoms, to provide a diagnosis.

- A fine pressure catheter is placed in the bladder through the urethra and a second catheter in the rectum, and the bladder is filled with warm saline while pressure recordings are made with the patient sitting on a commode that records leakage .
- The pressure generated by any contraction of the detrusor can be inferred by subtraction
(bladder pressure – abdominal pressure = detrusor pressure)
- During filling the patient is asked to declare the onset of bladder filling sensation (usually around 150 ml volume),

- a strong desire to void (around 350 ml and the onset of urgency (up to 500 ml, depending on bladder capacity)
- When urgency is reported (and functional bladder capacity is reached) filling is stopped and the patient performs various actions to provoke leakage and/or detrusor contractions (e.g. coughing, star jumps, listening to running water), before voiding while the pressure catheters remain in place.

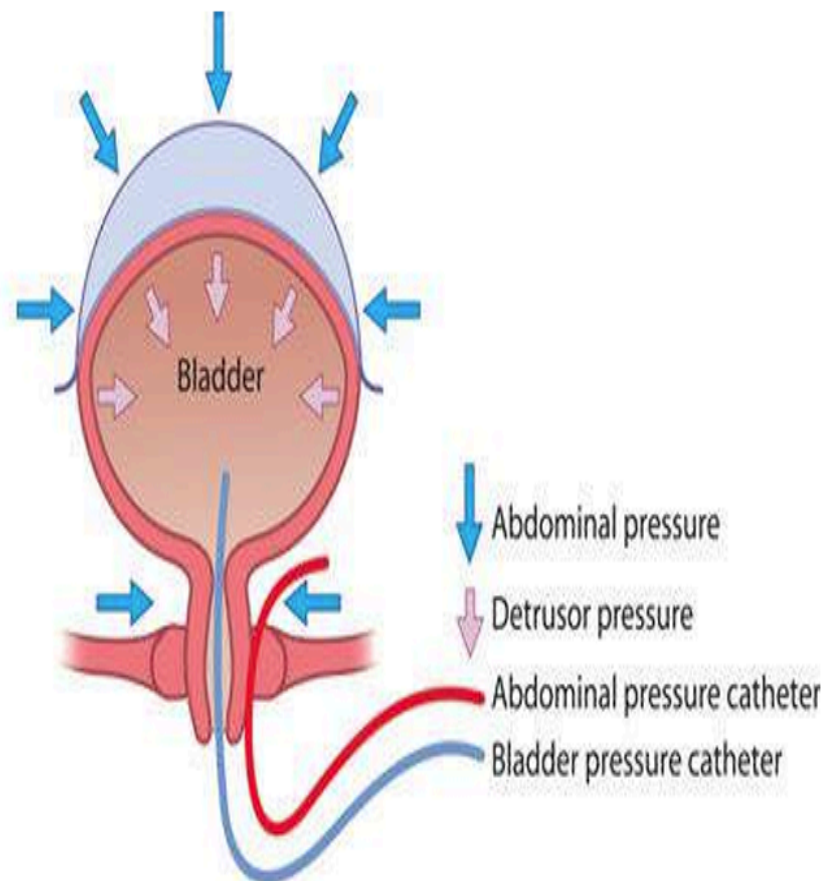
A urodynamic test will provide evidence of urethral sphincter weakness or DO as well as identifying normal or abnormal voiding function.

Urodynamic testing is reserved for patients who fail to improve with conservative measures, and also for patients with recurrent symptoms

or those with complex histories or who have undergone previous surgery for incontinence or prolapse .

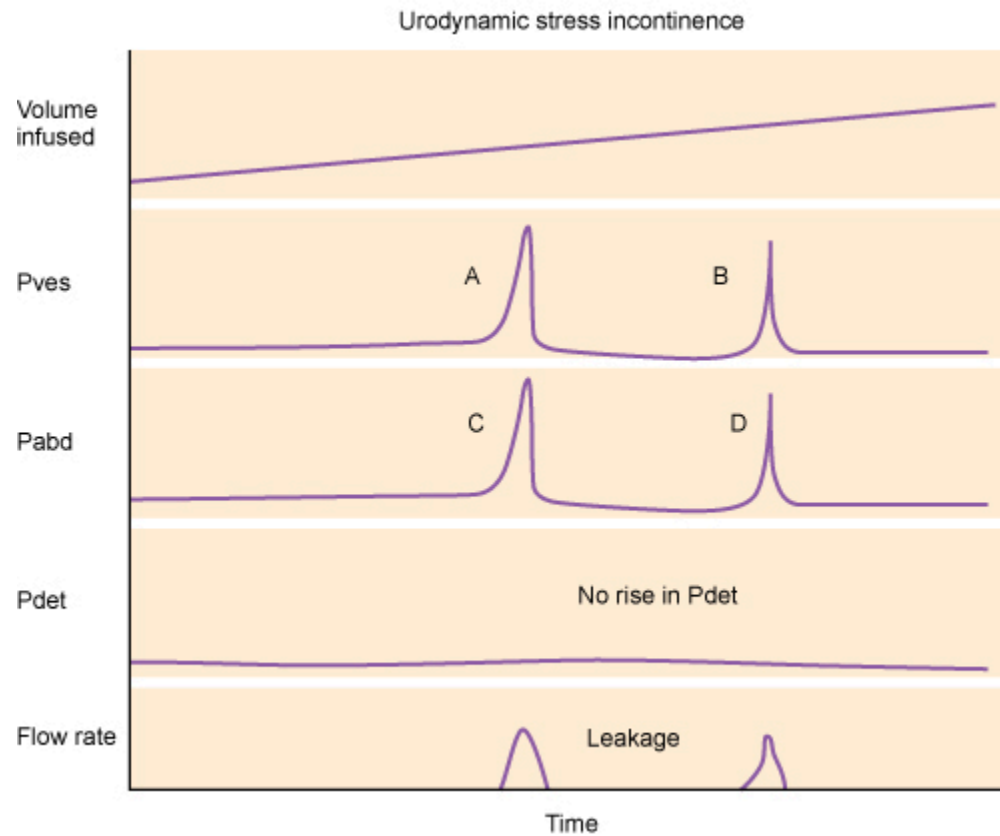
For women with OAB, many clinicians will initiate drug treatment before urodynamic testing , and only perform the investigation when considering second-line interventions

A



$$\text{Detrusor pressure} = \text{bladder pressure} - \text{abdominal pressure}$$

Urodynamic stress incontinence



Cystoscopy

Do not use cystoscopy in the initial assessment of women with UI alone.

Imaging

Do not use imaging (MRI, CT, X-ray) for the routine assessment of women with UI.

Do not use ultrasound other than for the assessment of residual urine volume.



Lifestyle interventions

A -Caffeine

Recommend a trial of caffeine reduction to women with OAB.

B -Fluid intake

Consider advising modification of high or low fluid intake in women with UI or OAB.

C- Weight

Advise women with UI or OAB who have a BMI greater than 30 to lose weight.

D -Physical therapies

Pelvic floor muscle training: Offer a trial of supervised pelvic floor muscle training of at least 3 months' duration as first-line treatment to women with stress or mixed UI. Pelvic floor muscle training programmes should comprise at least 8 contractions performed 3 times per day.

E- Continue an exercise programme if pelvic floor muscle training is beneficial.

- .

- Bladder training

Offer bladder training lasting for a minimum of 6 weeks as first-line treatment to women with urgency or mixed UI.

- Multicomponent behavioural therapy

If women do not achieve satisfactory benefit from bladder training programmes, the combination of an OAB drug with bladder training should be considered.



Products to prevent leakage

Do not use intravaginal and intraurethral devices for the routine management of UI in women. Do not advise women to consider such devices other than for occasional use when necessary to prevent leakage, for example during physical exercise.

Preventive use of conservative therapies : Offer pelvic floor muscle training to women in their first pregnancy as a preventive strategy for UI.



Pharmacological treatment

- General principles: when using OAB drugs When offering antimuscarinic drugs to treat OAB always take account of:
 - the woman's coexisting conditions (for example, poor bladder emptying)
 - use of other existing medication affecting the total anticholinergic load.
 - risk of adverse effects.

Pharmacological treatment continue

Before OAB drug treatment starts, discuss with women:
the likelihood of success and associated common adverse effects, and the frequency and route of administration, and that some adverse effects such as dry mouth and constipation. may indicate that treatment is starting to have an effect, and that they may not see the full benefits until they have been taking the treatment for 4 weeks.

Prescribe the lowest recommended dose when starting a new OAB drug treatment. If a woman's OAB drug treatment is effective and well-tolerated, do not change the dose or drug.



Offer one of the following choices first to women with OAB or mixed UI:

- Oxybutynin (immediate release), or
- Tolterodine (immediate release), or
- Darifenacin (once daily preparation).
- If the first treatment for OAB or mixed UI is not effective or well-tolerated, offer another drug with the lowest acquisition cost
- Offer a transdermal OAB drug to women unable to tolerate oral medication.
- . Do not offer oxybutynin (immediate release) to frail older women.



Desmopressin

The use of desmopressin may be considered specifically to reduce nocturia in women with UI or OAB who find it a troublesome symptom. Use particular caution in women with cystic fibrosis and avoid in those over 65 years with cardiovascular disease or hypertension



Duloxetine

Do not use duloxetine as a first-line treatment for women with predominant stress UI. Do not routinely offer duloxetine as a second-line treatment for women with stress UI, although it may be offered as second-line therapy if women prefer pharmacological to surgical treatment or are not suitable for surgical treatment. If duloxetine is prescribed, counsel women about its adverse effects.



Oestrogens

Do not offer systemic hormone replacement therapy for the treatment of UI.

Offer intravaginal oestrogens for the treatment of OAB symptoms in postmenopausal women with vaginal atrophy.



Surgical approaches for SUI

If conservative management for SUI has failed, offer:

Laparoscopic or open
colposuspension or autologous rectus fascial sling

a synthetic mid-urethral tape procedure, should be used only
for research purposes.



Intramural bulking agents

Consider intramural bulking agents (silicone, carbon-coated zirconium beads or hyaluronic acid/dextran copolymer) for the management of stress UI if conservative management has failed.

Women should be made aware that:

repeat injections may be needed to achieve efficacy

efficacy diminishes with time

efficacy is inferior to that of synthetic tapes or autologous rectus fascial slings



Pregnancy

- Episiotomy does not appear to protect the pelvic floor or prevent stress incontinence.
- Caesarean section is not completely protective against urinary and faecal incontinence.
- Postnatal pelvic-floor exercises are effective in reducing urinary incontinence.
- Antenatal pelvic-floor exercises are effective in risk groups.



Conclusion

- Stress incontinence is typically a result of weak urethral sphincter often as a consequence of childbirth.
- Detrusor overactivity causes urgency, frequency, and nocturia but not all patients will have leakage.
- History should include direct questions about faecal leakage or leakage during sexual intercourse.
- Clinical examination should exclude pelvic mass e.g. fibroid uterus and assess patient ability to contract her pelvic muscle and the strength of the contraction.



- Conservative treatment with fluid advice, pelvic floor exercises and bladder retraining is the first line treatment for all patient of incontinence or OAB and ideally should be delivered in primary care.
- Patient motivation is important in success of conservative treatment.
- Urodynamic investigation is not always necessary before commencing drug treatment or before surgery in women complaining only of stress incontinence.
- Surgery for stress incontinence is highly effective.
- Patient should be warned of the risk of voiding dysfunction, bladder injury and new OAB symptoms after surgery.